

Module Guide for Physiotherapy Companion

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April 12, 2026

1 Revision History

Date	Version	Notes
Jan 21, 2026	1.0	Initial documentation completed.
April 6, 2026	1.1	Feedback incorporated.

2 Reference Material

This section records information for easy reference.

2.1 Abbreviations and Acronyms

symbol	description
AC	Anticipated Change
DAG	Directed Acyclic Graph
M	Module
MG	Module Guide
OS	Operating System
R	Requirement
SC	Scientific Computing
SRS	Software Requirements Specification
Physiotherapy Companion	Explanation of program name
UC	Unlikely Change

Contents

1	Revision History	i
2	Reference Material	ii
2.1	Abbreviations and Acronyms	ii
3	Introduction	1
4	Anticipated and Unlikely Changes	2
4.1	Anticipated Changes	2
4.2	Unlikely Changes	3
5	Module Hierarchy	4
6	Connection Between Requirements and Design	5
7	Module Decomposition	6
7.1	Hardware Hiding Modules	6
7.2	Behaviour-Hiding Modules	7
7.2.1	Account Creation Module (M1)	7
7.2.2	Analysis Control Flow Module (M6)	7
7.2.3	Draw Module (M10)	8
7.2.4	User Interface Module (M11)	8
7.3	Software Decision Module	8
7.3.1	Valid User Module (M2)	8
7.3.2	User Account Module (M3)	9
7.3.3	Exercise Data Module (M4)	9
7.3.4	User Activity Module (M5)	10
7.3.5	Analyze Module (M7)	10
7.3.6	Feedback Module (M8)	10
7.3.7	Metrics Module (M9)	11
8	Traceability Matrix	11
9	Use Hierarchy Between Modules	13
10	User Interfaces	13
11	Design of Communication Protocols	13
12	Timeline	15

List of Tables

1	Module Hierarchy	5
2	Trace Between Requirements and Modules	12
3	Trace Between Anticipated Changes and Modules	13

List of Figures

1	Use hierarchy among modules	13
2	Modules with external communications	15

3 Introduction

Decomposing a system into modules is a commonly accepted approach to developing software. A module is a work assignment for a programmer or programming team (Parnas et al., 1984). We advocate a decomposition based on the principle of information hiding (Parnas, 1972). This principle supports design for change, because the “secrets” that each module hides represent likely future changes. Design for change is valuable in SC, where modifications are frequent, especially during initial development as the solution space is explored.

Our design follows the rules laid out by Parnas et al. (1984), as follows:

- System details that are likely to change independently should be the secrets of separate modules.
- Each data structure is implemented in only one module.
- Any other program that requires information stored in a module’s data structures must obtain it by calling access programs belonging to that module.

After completing the first stage of the design, the Software Requirements Specification (SRS), the Module Guide (MG) is developed (Parnas et al., 1984). The MG specifies the modular structure of the system and is intended to allow both designers and maintainers to easily identify the parts of the software. The potential readers of this document are as follows:

- New project members: This document can be a guide for a new project member to easily understand the overall structure and quickly find the relevant modules they are searching for.
- Maintainers: The hierarchical structure of the module guide improves the maintainers’ understanding when they need to make changes to the system. It is important for a maintainer to update the relevant sections of the document after changes have been made.
- Designers: Once the module guide has been written, it can be used to check for consistency, feasibility, and flexibility. Designers can verify the system in various ways, such as consistency among modules, feasibility of the decomposition, and flexibility of the design.

The rest of the document is organized as follows. Section 4 lists the anticipated and unlikely changes of the software requirements. Section 5 summarizes the module decomposition that was constructed according to the likely changes. Section 6 specifies the connections between the software requirements and the modules. Section 7 gives a detailed description of the modules. Section 8 includes two traceability matrices. One checks the completeness of the design against the requirements provided in the SRS. The other shows the relation between anticipated changes and the modules. Section 9 describes the use relation between modules.

4 Anticipated and Unlikely Changes

This section lists possible changes to the system. According to the likeliness of the change, the possible changes are classified into two categories. Anticipated changes are listed in Section 4.1, and unlikely changes are listed in Section 4.2.

4.1 Anticipated Changes

Anticipated changes are the source of the information that is to be hidden inside the modules. Ideally, changing one of the anticipated changes will only require changing the one module that hides the associated decision. The approach adapted here is called design for change.

AC1: The user interface of the application may be changed to include additional features for accessibility. Changes include:

1. How feedback is presented to the user as the exercise is performed (i.e. written, auditory, alerts, etc.).
2. How the insights are generated to the user on their account dashboard (i.e. displayed as a line graph, columns of data, does the user have the option to select from multiple views, etc.).

These changes would be reflected in the Feedback Module (M8) and the User Interface Module (M11). These modules handle the feedback generation from both captured metrics in the videos and the overall statistics of user activities, as well as the logic for displaying that information to the user. It is expected that these changes would increase clarity of stored data and data manipulation within the system, as well as increase patient confidence with better feedback and extendable exercise information.

AC2: The angles that are calculated to support different exercises. Currently, the application supports the 'single leg raise' exercises which is routinely used in rehabilitation for post-operative knee surgery. Future iterations of the application may include additional body parts and exercises.

This change would be reflected in the Exercise Data Module (M4), the Analyze Module (M7), and the Metrics Module (M9). The Exercise Data Module (M4) maintains a library of all the exercises assigned to each user, and it also facilitates the addition of exercises by physiotherapists to assigned patients and the modification of instructions, sets, and reps to assigned exercises. The Analyze Module (M7) calculates joint angles from landmark coordinates and would have to be modified to vary across different landmarks. IT also measures time-under-tension and exercise duration which are required for all the exercises in the system. The Metrics Module (M9) processes video frames to detect and track body landmarks using pose estimation, identifies key joints relevant to knee exercises (hip, knee, ankle, foot), extracts coordinate data for joint positions across video frames, and implements the core CV library integration (OpenCV, MediaPipe). This would also need to be modified to accept variable landmark selection by different exercises.

AC3: Changes to the feedback/guidance given. As more requirements are met, there should be changes to encapsulate the prompts required for the motion (i.e. 'keep the torso straight as this movement is performed'). Additionally, there may be alternate prompts with varying degrees of complexity, which relates to accessibility and the patients understanding of the exercise (i.e. 'during this motion you want to engage your quadriceps as you drive your toe towards the sky 8 inches off the ground'). These changes would be reflected in the Feedback Module (M8). This module generates real-time corrective feedback messages based on form analysis results, provides step-by-step exercise guidance, creates descriptive messages for form deviations specifying required adjustments, and triggers safety warnings for dangerous movements. It is expected that these changes would increase patient confidence and motivation to perform the exercises correctly, as well as increase the accessibility of the system to a wider range of patients with varying levels of understanding of the exercises.

4.2 Unlikely Changes

The module design should be as general as possible. However, a general system is more complex. Sometimes this complexity is not necessary. Fixing some design decisions at the system architecture stage can simplify the software design. If these decision should later need to be changed, then many parts of the design will potentially need to be modified. Hence, it is not intended that these decisions will be changed.

UC1: Input/Output devices (Input: Recording/Video Upload, Output: On-Screen Feedback/Corrections).

Modifying devices takes an extended amount of work to allow for different IPs for selected cameras, different formats are a possibility, as well as varying frame rates and possible latency on feedback generation; limitations in the video handling modules (M7-M10) and the Feedback Module (M8) are expected to be the most affected by this change.

UC2: The knee will remain modeled as a single-degree-of-freedom (1-DOF) hinge joint in the sagittal plane (flexion and extension only). No expansion to multi-DOF modelling or other joints is planned.

This design would require a data structure change in all the video handling modules (M7-M10) to support the additional landmarks and angles, as well as many additional changes to the Analyze Module (M7) to support more angles and form criteria.

UC3: Users will only perform physiotherapist-approved rehabilitation exercises. The system will not automatically generate or recommend new, unverified exercises.

This specifically has to do with health regulations and the system's inability to provide medical care advice to patients using the system.

UC4: Feedback will be limited to movement-based corrections and joint angle guidance. The system will not provide medical diagnosis or replace professional physiotherapy

assessment.

This was a difficulty faced with the latency on the system, a complete redesign of the front-end would be required in a python front-end language to support better real-time feedback it is estimated. However even this is not a guaranteed fix, so this excess amount of work for no guarantee of success is not worth the risk at this stage of development.

UC5: The system is not being built to integrate feedback from external sensors, wearable devices, and specialized specialized motion-capture equipment.

This ties back to the limitation and UC 1, as well as the fact that the system is being built to be accessible to patients at home with only a smartphone, so the addition of external sensors would limit the accessibility of the system and increase the cost for patients.

UC6: The scope will remain focused on knee rehabilitation. The system will not expand to whole-body tracking or unrelated physiotherapy movements.

This is primarily a scope issue, however you can see the implications of this change in the video handling modules (M7-M10) and the Analyze Module (M7) as per UC 2 and 3. They would need to be modified to support additional joints and angles, as well as the addition of new modules to support the new body parts and exercises.

5 Module Hierarchy

This section provides an overview of the module design. Modules are summarized in a hierarchy decomposed by secrets in Table 1. The modules listed below, which are leaves in the hierarchy tree, are the modules that will actually be implemented.

M1: Account Creation Module

M2: Valid User Module

M3: User Account Module

M4: Exercise Data Module

M5: User Activity Module

M6: Analysis Control Flow Module

M7: Analyze Module

M8: Feedback Module

M9: Metrics Module

M10: Draw Module

M11: UI Module

Level 1	Level 2
Hardware-Hiding Module	
	Account Creation Module
	Analysis Control Flow Module
Behaviour-Hiding Module	Draw Module
	User Interface Module
	Valid User Module
	User Account Module
	Exercise Data Module
Software Decision Module	User Activity Module
	Analyze Module
	Feedback Module
	Metrics Module

Table 1: Module Hierarchy

6 Connection Between Requirements and Design

The design of the system is intended to satisfy the requirements developed in the SRS. In this stage, the system is decomposed into modules. The connection between requirements and modules is listed in Table 2.

Design Decisions

Security To satisfy security and privacy requirements (FR21–FR23, SR-IR1, SR-PR1–2), the system implements end-to-end encryption for data at rest and in transit using industry-standard encryption protocols. User authentication employs session-based tokens with automatic timeout after 60 minutes of inactivity.

This functionality is included in both the Account Creation Module (M1) and the Valid User Module (M2) to ensure secure account creation and login processes. The User Account Module (M3) also incorporates role-based access control to restrict data access to authorized users.

Real-time Performance To meet the requirement for real-time feedback (FR13–14, PR-SL1–2), the computer vision processing pipeline is optimized to provide analysis results within 1 second of detecting form deviations. This is achieved through efficient pose estimation algorithms and selective processing of critical joint angles.

The Analysis Control Flow Module (M6), with assistance from the Analyze Module (M7), the Feedback Module (M8), the Metrics Module (M9), and the Draw Module

(M10), is designed to minimize latency and ensure timely feedback to users during exercise sessions. All video operations are performed in a flow between these modules, each controlling a finer band of information handed from the previous module.

Modular Exercise Library To support extensibility (PR-CR1, PR-SER1), the exercise library is designed as a modular system where new exercises and joint types can be added without modifying core analysis logic. Each exercise type has its own parameter set defining acceptable ranges and form criteria.

This functionality is provided by the Exercise Data Module (M4), which maintains a library of all the exercises assigned to each user. It also facilitates the addition of exercises by physiotherapists to assigned patients and the modification of instructions, sets, and reps to assigned exercises. All of this information is stored in the database, to ensure consistency across devices and sessions.

Two-Camera View Analysis To address the risk of 3D motion measurement with 2D cameras, the system supports analysis from the sagittal view as primary. The analysis algorithms are designed to work with a single camera from nearly any angle, supporting the data gathering of angles across the knee from almost any perspective.

As this functionality is related to video, it again falls to the video handling modules (M7-10). The dot product of the vectors created between landmarks placed on the video gets a little less accurate at certain perspectives, but most are still accurate enough to support the functionality of our system and uphold requirements.

7 Module Decomposition

Modules are decomposed according to the principle of “information hiding.” The *Secrets* field in a module decomposition is a brief statement of the design decision hidden by the module. The *Services* field specifies *what* the module will do without documenting *how* to do it. If the entry is *OS*, this means that the module is provided by the operating system or by standard programming language libraries. *Physiotherapy Companion* means the module will be implemented by the Physiotherapy Companion software. Only the leaf modules in the hierarchy have to be implemented.

7.1 Hardware Hiding Modules

Secrets: The data structure and algorithm used to implement the virtual hardware.

Services: Serves as a virtual hardware used by the rest of the system. This module provides the interface between the hardware and the software, including camera access, screen display, audio output, and device sensors. The system can use it to capture video, display outputs, or accept touch inputs.

Implemented By: N/A (OS and standard libraries through React Native)

7.2 Behaviour-Hiding Modules

Secrets: The contents of the required behaviours.

Services: Includes programs that provide externally visible behaviour of the system as specified in the software requirements specification (SRS) documents. This module serves as a communication layer between the hardware-hiding module and the software decision module. The programs in this module will need to change if there are changes in the SRS.

Implemented By: Physiotherapy Companion

7.2.1 Account Creation Module (M1)

Secrets: The logic for creating and storing both PTs and patient accounts, and creating physiotherapist invite codes.

Services: Handles new user account creations for both patients and PTs with input information. Also generates unique invite codes for PT accounts upon creation for linking patient accounts. Stores user account information securely in a remote database.

Implemented By: Physiotherapy Companion

Type of Module: Library Module

Related Requirements: FR1, FR2

7.2.2 Analysis Control Flow Module (M6)

Secrets: The implementation of video recording, including camera configuration, frame rate management, and recording controls.

Services: Captures video of users performing exercises at minimum 720p resolution and 30 fps. Performs pre-capture environment checks (lighting, framing). Provides recording controls (start, stop, retry). Ensures video meets minimum quality standards before proceeding to analysis.

Implemented By: Physiotherapy Companion

Type of Module: Abstract Object

Related Requirements: FR7, FR9, FR13, FR15, FR26, UHR-EOU3

7.2.3 Draw Module (M10)

Secrets: The graphical representation of landmarks and objects of interest within the video provided by the user.

Services: Graphically shows the user the system's understanding of their movements by highlighting landmarks such as their ankle, knee, and hip. Additionally, draws lines between these landmarks to communicate how the system interprets limb movement.

Implemented By: Physiotherapy Companion

Type of Module: Library

Related Requirements: FR5, FR14, LFR-ST1, UHR-EOU2, UHR-EOU5, UHR-EOU6

7.2.4 User Interface Module (M11)

Secrets: The user interface layout and interaction logic for patient and physiotherapist dashboards.

Services: Displays patient exercise history, progress charts, and upcoming exercises. Shows PT patient list with status indicators and alerts. Provides access to video review and report viewing. Enables PT comments on patient sessions. Presents analytics and visualizations of exercise data.

Implemented By: Physiotherapy Companion

Type of Module: Abstract Object

Related Requirements: FR18, FR19, FR24, FR25

7.3 Software Decision Module

Secrets: The design decision based on mathematical theorems, physical facts, or programming considerations. The secrets of this module are not described in the SRS.

Services: Includes data structure and algorithms used in the system that do not provide direct interaction with the user.

Implemented By: Physiotherapy Companion

7.3.1 Valid User Module (M2)

Secrets: The validation and authorization logic for all user credentials (patient and PT), invite codes, and physiotherapist identification numbers.

Services: Checks input user information to ensure unique entries into the remote database. Ensures valid invite code during patient account creation to link new account to relevant PT account. Ensures physiotherapist identification numbers are valid through the publicly accessible records during PT account creation.

Implemented By: Physiotherapy Companion

Type of Module: Library Module

Related Requirements: FR1, FR3

7.3.2 User Account Module (M3)

Secrets: The authorization logic for existing user actions, and the retrieval and modification logic for user information in the remote database.

Services: User logins (both patient and PT) authorized by checking the remote database for matching email and password. Enables physiotherapists to delete linked patient accounts for housekeeping and service limitation purposes. Additionally, retrieves all relevant user account information from the remote database for other module usage.

Implemented By: Physiotherapy Companion

Type of Module: Data Access Module

Related Requirements: FR1, FR2, FR4, FR18, FR21, FR22, FR23, SR-AR1, SR-AR3, SR-IR1, SR-PR1, SR-PR2

7.3.3 Exercise Data Module (M4)

Secrets: The structure and storage logic for user relevant exercise data.

Services: Provides exercise information to users on demand. Maintains a library of knee recovery relevant exercises along with exercise instructions. Allows user specific modification of prescribed exercises by linked PT account.

Implemented By: Physiotherapy Companion

Type of Module: Abstract Object

Related Requirements: FR5, FR6, FR13, FR21, MS-AD1, SR-IR1, SR-PR1

7.3.4 User Activity Module (M5)

Secrets: The structure and storage logic for users historic activity data.

Services: Formats, links, and stores user activity data in the remote database in a table for historic activity summaries. Formats input from the Analyze Module.

Implemented By: Physiotherapy Companion

Type of Module: Abstract Object

Related Requirements: FR21, FR25, SR-IR1

7.3.5 Analyze Module (M7)

Secrets: The mathematical models and algorithms for calculating joint angles, range of motion, and determining form correctness based on exercise parameters. Also the mechanism for storing video files through local caching.

Services: Calculates joint angles from landmark coordinates with $\pm 5^\circ$ accuracy. Measures range of motion with $\pm 10\%$ accuracy. Compares measured values against exercise-specific acceptable ranges. Detects form deviations and classifies severity. Identifies dangerous movements requiring immediate halt. Measures time-under-tension and exercise duration. Also stores recorded exercise videos securely (encrypted at rest per SR-IR1).

Implemented By: Physiotherapy Companion

Type of Module: Abstract Data Type

Related Requirements: FR12, FR13, FR14, FR21, SR-IR1, PR-RFT2, PR-PAR1, PR-PAR2, PR-SCR1, MS-MNT1

7.3.6 Feedback Module (M8)

Secrets: The rules and logic for converting analysis results into user-friendly feedback messages and instructions.

Services: Generates real-time corrective feedback messages based on form analysis results. Provides step-by-step exercise guidance. Creates descriptive messages for form deviations specifying required adjustments. Triggers safety warnings for dangerous movements.

Implemented By: Physiotherapy Companion

Type of Module: Library

Related Requirements: FR13, FR14, FR16, PR-SCR1

7.3.7 Metrics Module (M9)

Secrets: The specific computer vision algorithms and implementation details for pose detection, joint tracking, and body landmark identification.

Services: Processes video frames to detect and track body landmarks using pose estimation. Identifies key joints relevant to knee exercises (hip, knee, ankle, foot). Extracts coordinate data for joint positions across video frames. Handles varying lighting conditions and camera angles. Implements the core CV library integration (OpenCV, MediaPipe).

Implemented By: Physiotherapy Companion

Type of Module: Abstract Data Type

Related Requirements: FR8, FR12

8 Traceability Matrix

This section shows two traceability matrices: between the modules and the requirements and between the modules and the anticipated changes.

Req.	Modules
FR1	M1, M2, M3
FR2	M1, M3
FR3	M2
FR4	M3
FR5	M4, M10
FR6	M4, M11
FR7	M6
FR8	M6, M9
FR9	M6, M11
FR10	M11
FR11	M11
FR12	M7, M9
FR13	M4, M6, M7, M8
FR14	M7, M8, M10
FR15	M6
FR16	M8
FR17	M11

FR18	M3, M11
FR19	M11
FR20	M11
FR21	M3, M4, M5, M7
FR22	M3, M11
FR23	M3
FR24	M11
FR25	M5, M11
FR26	M6
FR27	M11
LFR-ST1	M10
MS-AD1	M4
MS-MNT1	M7
PR-PAR1	M7
PR-PAR2	M7
PR-RFT2	M7
PR-SCR1	M7, M8
SR-AR1	M3
SR-AR2	M2
SR-AR3	M3, M11
SR-IR1	M3, M4, M5, M7
SR-PR1	M3, M4
SR-PR2	M3
UHR-EOU2	M10
UHR-EOU3	M6
UHR-EOU5	M10
UHR-EOU6	M10

Table 2: Trace Between Requirements and Modules

AC	Modules
AC1	M6, M8, M11
AC2	M4, M5, M6, M7, M9, M11
AC3	M6, M7, M8, M9

Table 3: Trace Between Anticipated Changes and Modules

9 Use Hierarchy Between Modules

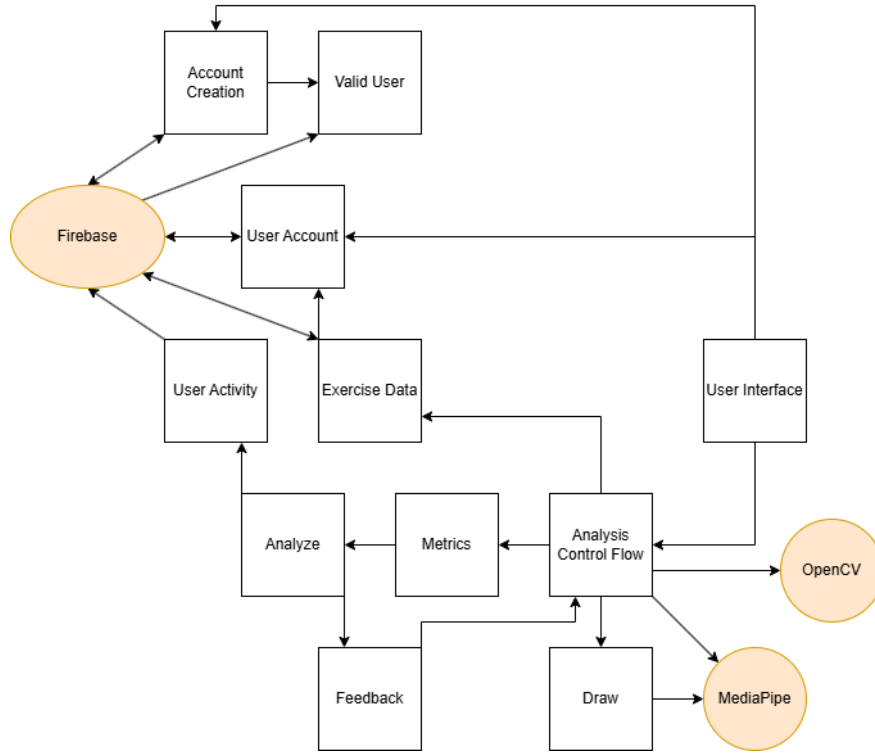


Figure 1: Use hierarchy among modules

10 User Interfaces

The proposed user interface for the system will attempt to model [this Figma prototype](#).

11 Design of Communication Protocols

Hardware intake is limited to the video feed. This feed, provided by the respective system's hardware layer, is requested and used by the Analysis Control Flow Module. This

module also overlays respective data returned to it from the Metrics Module. Additionally, the metrics module makes API calls to OpenCV and MediaPipe API to determine necessary information such as joint angle, movement speed, exercise duration, and number of repetitions.

The Metrics Module also passes this relevant information to the Analyze Module in real-time where the numerical returns are inspected to ensure exercises are being performed within bounds. The Analyze Module makes necessary calls to the Feedback Module if the user should be told to change form, speed, or other such exercise relevant information. The Analyze Module also passes generalized exercise information to the User Activity Module for storing exercise overviews tied to user accounts.

The Account Creation, Valid User, User Account, User Activity, and Exercise Data Modules interact with the external database hosted by Firebase.

The Account Creation Module uploads new account information to a table kept for user information which includes details relevant for login and account validation.

The Valid User Module checks the user information table to ensure an account doesn't exist with identical information and that the entered invitation code is valid to an existing physiotherapist account.

The User Account Module both uploads and retrieves information from the user information table for usage on the main account page and for linking relevant user activities and exercise data.

The User Activity Module uploads information to an activity tracking table in the database. This table is used to show both the user and the linked physiotherapist the results and overall progress on prescribed exercises.

The Exercise Data Module both uploads and retrieves information from a table for specific exercise alterations. This information is editable by physiotherapist accounts and is referenced by the Analysis Control Flow Module during exercise recording.

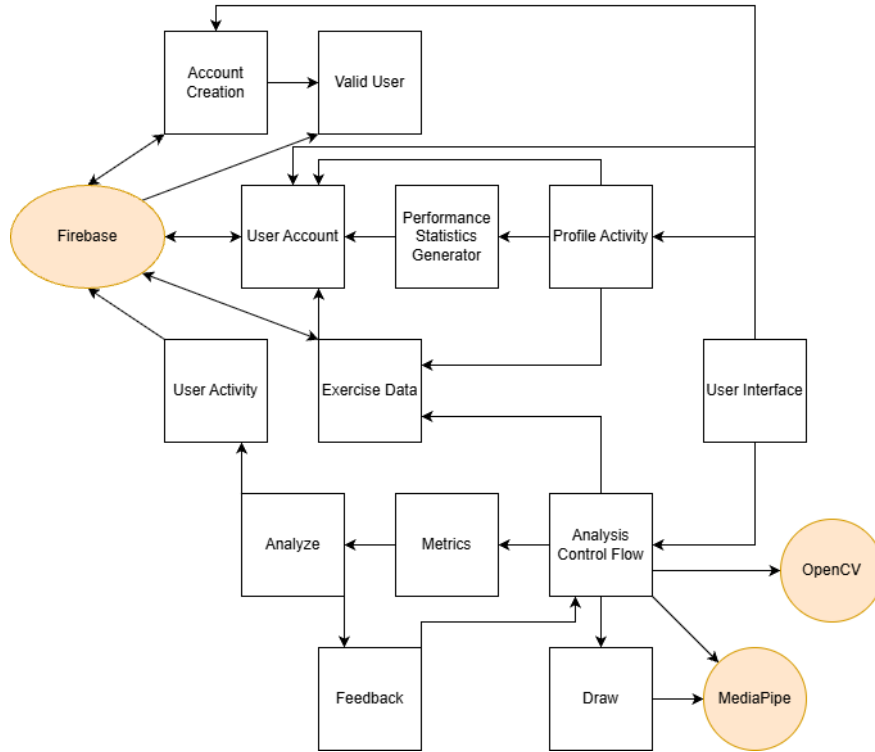


Figure 2: Modules with external communications

12 Timeline

For the remainder of the term, the project timeline will follow the official course deliverables and our internal task assignments. A detailed task breakdown is maintained on our [GitHub Project Board](#). The high-level schedule is listed below:

- **November 14th, 2025: Analysis Control Flow Module** (Responsible: Matthew, Vaisnavi)
 - Video recording capabilities fully implemented and ready for POC.
- **November 14th, 2025: Draw Module** (Responsible: Matthew)
 - Landmark placement and line drawing capabilities fully implemented and ready for POC.
- **November 14th, 2025: Metrics Module** (Responsible: Matthew, Vaisnavi)
 - Numerical data extraction from video feed fully implemented and ready for POC.
- **November 14th, 2025: Analyze Module** (Responsible: Vaisnavi)

- Exercise analysis, feedback generation, and data parsing capabilities fully implemented and ready for POC.
- **November 16th, 2025: POC Testing** (Responsible: Entire Team)
 - Run throughs and debugging of the POC to ensure everything is working as expected before the POC demonstration.
- **During November, 2025: Design Doc Rev0 and further development** (Responsible: Eman, Cieran, Maham)
 - Work on the remaining modules (see dates below), and refine design documents from -1 to 0 for deliverable in January.
- **January 23rd, 2026: Design Doc Rev0** (Responsible: Entire Team)
 - Complete MG and MIS documents with all modules fully described and connected to requirements, and with the use hierarchy fully defined.
- **January 26th, 2026: User Account Module** (Responsible: Maham)
 - Complete implementation of account data manipulation capabilities, including user login/logout, session management, and retrieval of user information for other module usage.
- **January 26th, 2026: User Creation Module** (Responsible: Maham)
 - Complete implementation of account creation capabilities, including user input handling and database communication.
- **January 27th, 2026: UI Module** (Responsible: Vaisnavi)
 - React frontend development of the user interface with basic navigational capabilities, and respective places for linking backend data to be displayed.
- **January 27th, 2026: UI Module** (Responsible: Vaisnavi)
 - React frontend development of the user interface with basic navigational capabilities, and respective places for linking backend data to be displayed.
- **January 28th, 2026: User Activity Module** (Responsible: Cieran)
 - Complete implementation of user activity tracking capabilities, including logging and recalling of user activities from the database.
- **January 29th, 2026: Exercise Data Module** (Responsible: Eman)
 - Complete implementation of exercise data handling capabilities, including data collection, storage, and retrieval of exercise instructions.

- **January 30th, 2026: Valid User Module** (Responsible: Eman)
 - User validation logic implemented, including checks for unique user information and valid invite codes during account creation.
- **January 30th, 2026: UI Module Finalization** (Responsible: Cieran)
 - Complete implementation of dashboard, exercise, progress, and profile pages with database stored information, linked to backend modules.
- **January 26th, 2026: Rev0 Testing** (Responsible: Entire Team)
 - Runthroughs and debugging of the entire system to ensure everything is working as expected before the Rev0 demonstration.
- **February 5th, 2026: Rev0 Demonstration** (Responsible: Entire Team)
 - Complete implementation expected for display of all modules and functionalities to be demonstrated.
- **March 27th, 2026: Module Updates** (Responsible: Entire Team)
 - Updates from Rev0 to Rev1 based on Professor and TA feedback, including improvements to the UI, analysis capabilities, and overall system stability.
- **March 28th, 2026: Final Demonstration** (Responsible: Maham)
 - Demonstration to Professor and TA of the final system with all modules fully implemented and working together to satisfy the requirements. No further changes to implementation expected beyond this point.
- **April 6th, 2026: Final Documentation** (Responsible: Maham)
 - Final update and changes to this document and all related modules and design documents based on feedback from the final demonstration, and submission of all final documentation to the course repository.

References

- David L. Parnas. On the criteria to be used in decomposing systems into modules. *Comm. ACM*, 15(2):1053–1058, December 1972.
- D.L. Parnas, P.C. Clement, and D. M. Weiss. The modular structure of complex systems. In *International Conference on Software Engineering*, pages 408–419, 1984.